

Lab 02 - System Request

Project: Library Management System (LMS)

Prepared by: Omar Ayman Shebl Motawea, Gamal Hamed Gamal Aldin, Michael Erian Ayad sadhom

Date: October 2026

1. Project Sponsor

Eng. Eman Abdel Raoof, Instructor of Systems Analysis.

2. Business Need

The library currently depends on paper registers, index cards and spreadsheets. Searching for a book takes a long time, borrowing and returning are recorded by hand, and it is hard to know who has which book or which books are overdue. Fines are calculated manually, books get lost, and the librarian has no easy way to see reports.

The library needs a web-based system that keeps all books and members in one place, records borrowing and returning, and tracks overdue books and fines automatically.

3. Business Requirements

1. The librarian can add, edit and delete books (title, author, category, number of copies).
2. Members can register / log in and search for books by title, author or category.
3. The system records borrowing and returning of books with due dates.
4. The system detects overdue books and calculates fines automatically.
5. Members can reserve a book that is currently borrowed.
6. Admin can manage users and roles (admin, librarian, member).
7. The librarian can view reports (borrowed books, overdue books, popular books) and export them to Excel.
8. The system is secure (authentication and role-based access).

4. Business Value

Tangible: saves staff time in record keeping and searching; removes paper, registers and printing; reduces lost books; faster check-in and check-out.

Intangible: fewer record errors; faster access to books; better satisfaction for members; clear statistics for decision making.

5. Special Issues and Constraints

- **Time:** limited by the course schedule.
- **Budget:** low; use open-source tools and free hosting where possible.
- **Technology:** web-based, works on desktop and mobile browsers.
- **Risks:** staff resistance to change, wrong data entry when moving old records, privacy of member data.
- **Out of scope:** e-books, online payment of fines, RFID gates.

6. Converting Requirements into Project Tasks

Business requirement	Project task (Trello card)	Lab
Need and value are clear	Write System Request	02
Is the project worth it	Feasibility Study	03-04
Choose how to work	Choose methodology	03
Who does what	Staff plan and structure	05
Plan the work	WBS and timeline	06
Requirements 1-8	Requirements specification	07
Current vs proposed workflow	DFD As-Is and To-Be	08
System behavior	UML diagrams	09
Screens (login, search, borrow)	UI mockups	10
Deliver everything	Final documentation	11

7. Project Board

Trello board: <https://trello.com/invite/b/6ac7af7f3e9c3f65af5c13f8/ATTI10d6e06498620144829808d1c4dbb462CCB9D800/library-management-system>

Screenshot:

